

Abstract

This project analyzes the factors influencing singles tennis match outcomes on hard court and evaluates machine learning models for match prediction using ATP and WTA US Open dataset (2019-2024). A set of difference-based features are engineered to capture relative player performance, including rally consistency, serve performance, return performance and performance under pressure. Logistic Regression, Random Forest, and XGBoost models are trained and compared with cross validation and test F1 score.

All models achieved strong performance, with F1 score above 0.84 for ATP and 0.90 for WTA matches. Logistic Regression performed well, while tuned Random Forest and tuned XGBoost have the best results for ATP and WTA datasets respectively. Feature importance analysis shows that winner rate and unforced errors are the primary factors for match outcome. Break point performance is particularly important in WTA matches. Net point performance is particularly important in ATP matches. These findings prove the importance of rally consistency, controlled aggression, and performance under pressure for amateur players.

Introduction

A tennis match's outcome is influenced by many factors. The two head-to-head players' techniques, rankings, strategies to approach the game, how they handle the pressure situations, all contribute to the uncertainty of a match's result. There're already lots of models to predict match results or even point winners for professional tennis matches. Most of the time a match's result is decided largely by the two players' rankings and their Elo ratings (which can reflect a player's comprehensive strength).

This project will take a different perspective, focus on the match statistics factors, identify the most important factors for winning a singles tennis match, for example, does rally length matter for the match result, do you win by more winners or less unforced errors, is improving serve quality or avoid double fault more important?

The main goal of this project is to answer one question: what the most important factors for winning singles tennis matches at hard court. To answer this question, first we need to aggregate the points data into matches, and get match level dataset; then do correlation analysis about what statistic factors effect most about winning a match; then we build multiple machine learning models to predict match results based on those statistic factors, get feature importance, and finally derive the most important factors.

This analysis can provide training and game strategy insights for recreational players playing in local community leagues. It also only focusses on the hard court matches since that's where amateur players usually play on here in the United States.

Data sources

In tennis world there are four most significant tournaments each year, Austrian Open, French Open, Wimbledon Open, and US Open, which are called Grand Slam. Among them, Austrian Open and US Open are hard court tournaments. For the data source, I use the point level

datasets at [Jeff Sackmann's github tennis_slam_pointbypoint repository](#), since they give more detailed match statistics.

The Austrian Open datasets are incomplete. Thus, this project only use data from US Open. I choose the year 2019-2024 due to three reasons. First, recent years have more complete datasets; Second, for the earlier years, tennis matches were dominated by the 'Big Three' players, so it's not quite good for a more universal reference; Third, each dataset has over 48k data points. The yearly dataset is sufficient for this project.

I combine the 2019-2024 US Open pointbypoint singles match dataset into one dataset, which is defined as 'all_dfs'. It has 65 variables, 11 of them are empty. So, there is 54 variables in the raw dataset, including the information and results of a particular point, and the performance of both players (noted by P1 and P2). Some critical variables are explained as following:

match_id: which match this point belongs to; PointWinner: which player wins the point (1 indicates player1 wins; 2 indicates player2 wins); PointServer/ServeIndicator: which player serves this point; Speed_MPH: speed of the serve; ServeNumber: whether it's the first or second serve or double fault (0 indicates a double fault, 1 indicates the first serve, 2 indicates the second serve); ServeWidth: the serve placement (B indicates Body; C indicates Center; W indicates Wide); ServeDepth: the depth of a serve (CTL indicates a deep serve; NCTL indicates a not so deep serve); ReturnDepth: the depth of a return (D indicates it's a deep return; ND indicates it's a not deep return); P1PointsWon/P2PointsWon: the total points P1/P2 won until this point in the current match; P1Ace/P2Ace: whether P1/P2 serves an ace in this point; P1Winner/P2Winner: whether P1/P2 hits a winner in this point; The same as P1DoubleFault, P2DoubleFault, P1UnfErr, P2UnfErr, P1NetPoint, P2NetPoint, P1NetPointWon, P2NetPointWon, P1BreakPoint, P2BreakPoint, P1BreakPointWon, P2BreakPointWon, which indicate a player's technique performance; P1DistanceRun/P2DistanceRun: the total distance Player1/Player2 run before finishing this point; RallyCount: the total shots hit in this point.

More glossary of tennis terms can be found here:

https://en.wikipedia.org/wiki/Glossary_of_tennis_terms

Proposed Methodology

1.Data Engineering

First, I inspected and preprocess the raw point-level tennis data to ensure data consistency and completeness. The redundant and undefined variables (eg, duplicate speed metrics and 'History' field) were removed. Missing and invalid values were handled with domain specific logic. For example, RallyCount value equals to 0, which is invalid in the context of tennis, were treated as missing values, and imputed with a hierarchical strategy. Categorical variables (e.g., 'ServeWidth', 'ServeDepth', 'ReturnDepth') were not imputed at the point level, but were aggregated at the match level directly to preserve the distributional characteristics.

Second, I did match-level aggregation, transforming the point-level data(all_df) into a match-level dataset('match_df'). There're 49 variables in total, which includes performance metrics (e.g., aces, winners, unforced errors), efficiency statistics (e.g., net points won, break points

won), and match info (e.g., total points played, serve speed). The aggregation was performed separately for each player.

Third, I inspected the match-level dataset. Matches that fail the tennis-context validity constraints, for example, if fewer than 48 points, were removed. Continuous variables that equal to 0 (e.g., distance run, serve speed) were treated as missing and imputed using the mean substitution. I also derived each player's percentage statistics and get the difference statistics(e.g., 'winner_rate_diff', 'ue_rate_diff', 'ace_rate_diff'), since tennis is a relative performance sport. We can only make predictions based on the difference stats, not using the original absolute stats.

I combine the difference stats, core stats of a match, serve stats and absolute stats we already have, generate an ultimate dataset for analysis, which is called 'analysis_df'.

After all the processing, I make two extra engineering. First, I changed winner labels (1 remains 1, 2 is turned to 0) for the later classification. match_winner is the target classification predictor. Second, according to the match-level information (also from the same Github data source), I added another column named 'tour', which labels a match as 'atp' or 'wta'. (ATP indicates men's professional tennis, WTA indicates women's professional tennis)

The details of this data engineering process are described in the Appendix.

2. Models to predict the match results

In this project, we use the performance difference stats to predict whether a player win or lose a match. the target response is match_winner in the dataset. The most related features including winner_rate_diff, ue_rate_diff, break_Win_pct_diff, net_win_pct_diff, ace_rate_diff, df_rate_diff, returndepth_nd_diff are predictors according to the eda& correlation analysis results. Since features importance across ATP and WTA tours are different, I end up building models for them separately.

The methods used here are logistic regression, random forest and XGBoost since it's a classification problem. First, we split the total dataset into train (80%) and test (20%) dataset and use 5-fold cross validation on the training dataset to select the best model, then evaluate the model on test dataset. For the evaluation and comparison, we use accuracy and F1 score (more robust than accuracy) as the performance metrics.

Logistic regression: it captures the linear relationships between my predictors and response, provide interpretable results, and usually have a decent performance. It's the baseline model here. It is a linear combination of features, so feature scaling was applied for it to ensure the comparability of coefficients.

Random forest: it captures the nonlinear relationships and has more robust results. Here we use the default random forest classifier from sklearn first, and then perform the parameter tuning by using Grid Search. The parameter tuning grid include n_estimators(default =100), max_depth(default=None), min_sample_split(default=2), min_samples_leaf(default=1) and max_features(default='sqrt') . There are around 600 training datapoints and 7 predictors in each of the tour, considering it's a small dataset, the range value of n_estimators is [100, 200, 300, 500]; for max_depth, the range value is [None, 3, 5, 7];for min_samples_split, the range value is

[2, 5, 10, 15]; for min_samples_leaf, the range value is [1,2,4,6]; for max_features, the range value is [None, 'sqrt', 'log2']

XGBoost: it often has the best performance and can capture feature importance very well. We use the default xgboost classifier from sklearn first, then perform the parameter tuning by using Random Search. The parameter tuning grid include n_estimators(default =1.0), max_depth(default=3), learning_rate(default=0.1), and subsample(default=1.0). The range value of n_estimators is [50, 100, 200]; Here we use fewer trees than random forest in case of overfitting. For max_depth, the range value is [2, 3, 4, 5]; for learning_rate, the range value is [0.05, 0.1, 0.2]; for subsample, the range value is [0.8, 1.0].

In the end, we compare the f1 score and accuracy of all the models and get feature importance of the best model. From those features, we derive the insights of proper tennis training focus.

The whole analysis uses Google colab and python.

Analysis and Results

1. EDA & Correlation analysis

1.1 Overall Distributions

The analysis dataset has 60 variables and 1498 matches in total. All the missing values and wrong records have been processed previously when forming the analysis dataset.

For a match, the points played is between 52 to 398 points with a mean of 184. The sets played is either 2, 3, 4, or 5 sets. With a mean of 3 sets. The rallies count (shots played) is between 119 to 2075 with a mean of 673.

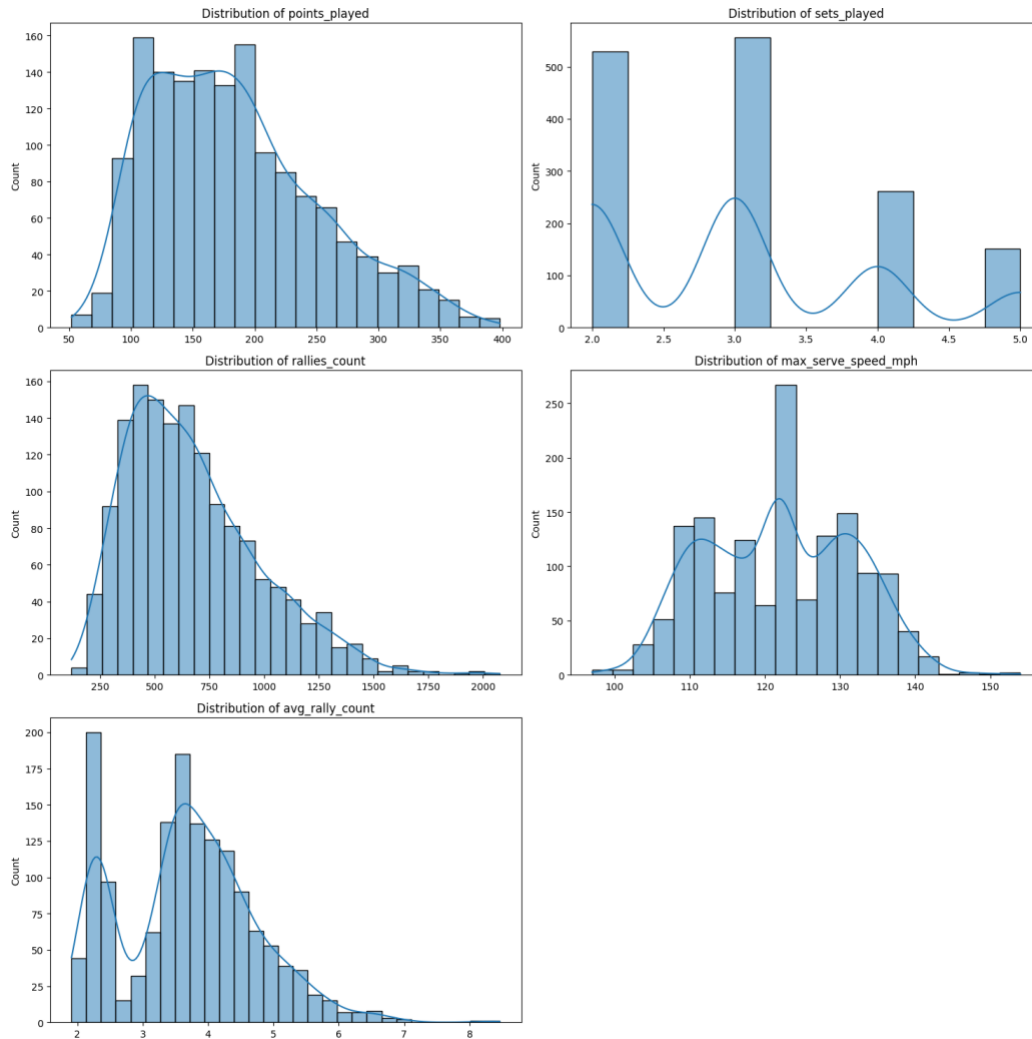


Figure1: distributions of match features

The performance difference stats, such as winner rate difference between two players, unforced error rate difference between two players.... all have an approximately normal distribution (see Appendix-B: supplement plots)

Since tennis is a comparative sport. Here we focus on match core info and difference stats.

2.2 ATP and WTA difference

From this boxplot, we can see ATP and WTA matches have significant difference. ATP matches have more points played, more sets played, more total shots played (due to long sets), yet the average shots played in a point is less. It also has much larger max serve speed.

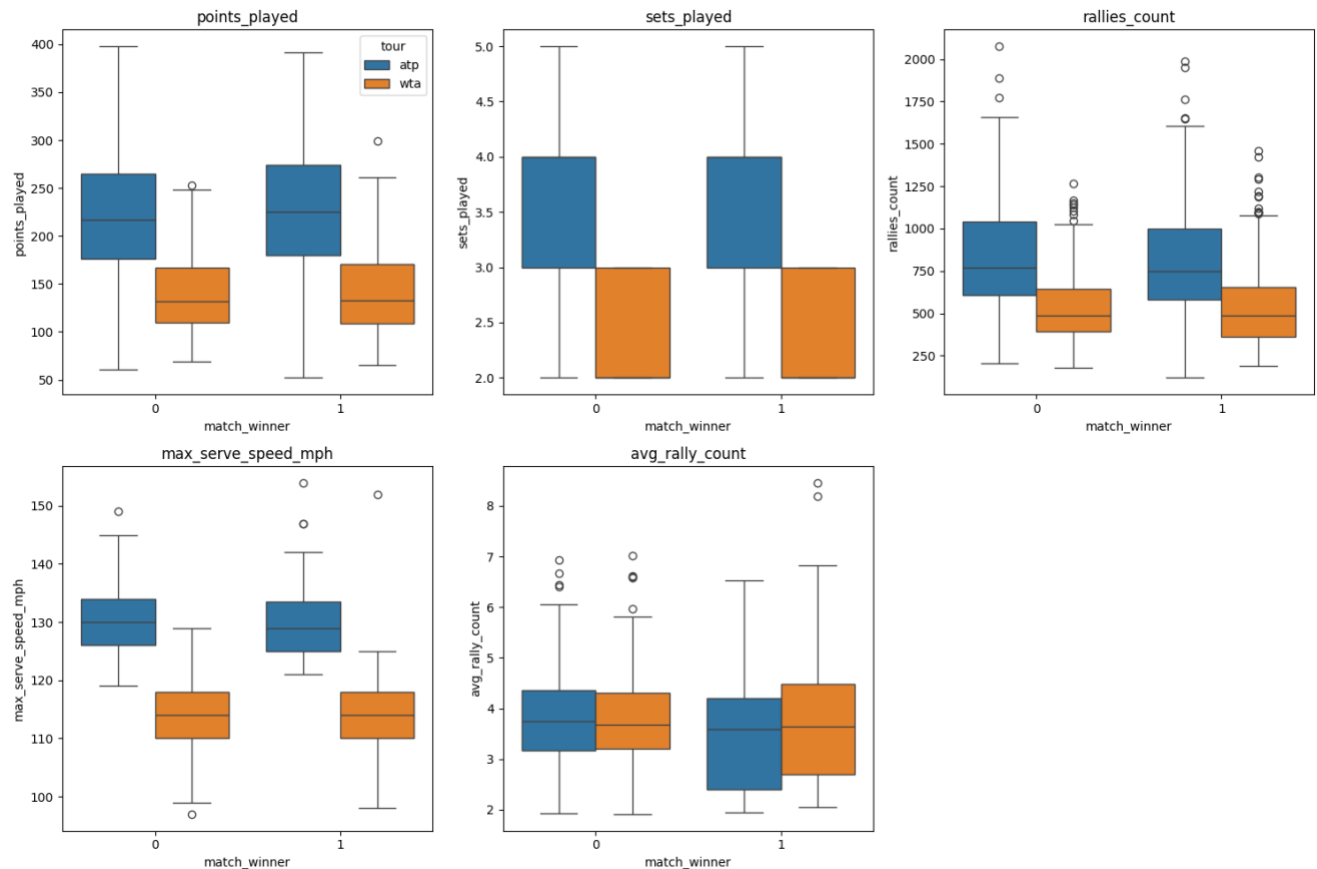


Figure2: boxplot of match features in both ATP and WTA

It shows that we'd better separate the ATP and WTA data points and do the model building separately for them. The performance difference stats' boxplot also supports this observation (see Appendix-B: supplement plots)

2.3 correlation matrix analysis

From the feature correlation matrix with match_winner for ATP and WTA separately(see Appendix), we have two conclusions: 1.the most correlated features for both ATP and WTA are the same, such as points_win_pct_diff, winner_rate_diff, ue_rate_diff, break_Win_pct_diff...; 2. Those features have different impact on ATP and AWTa. For example, net_win_pact_diff has stronger impact on ATP (0.48) than WTA (0.30), break_win_pct_diff has stronger impact on WTA (0.48) than ATP (0.42).

For machine learning models, we remove points_win_pct_diff from the feature set because it's highly correlated with match_winner(around 0.8), and it is a result feature like the match_winner, not a performance specific feature.

The final feature set we choose for the machine learning model are as following: winner_rate_diff, ue_rate_diff, break_Win_pct_diff, net_win_pct_diff, ace_rate_diff, df_rate_diff, returndepth_nd_diff. We'll use the same feature set to build models separately for ATP and WTA data.

2. Machine Learning Results

2.1 Model Performance

1. ATP dataset

The performance of all machine learning models on the ATP dataset is summarized in the Table1. Logistic Regression, Random Forest, and XGBoost all achieve strong results, with relatively small differences across evaluation metrics (Note: corss-validation accuracy for tuned models is not reported, as f1 score is used as the primary optimization metric during parameter tuning)

As the baseline model here, Logistic Regression has the highest cross-validation result (accuracy=0.900, f1=0.890), indicating that the relationships between the features and match outcome are mostly linear. But it's test performance(f1=0.841) is slightly lower than the ensemble methods.

Random Forest has a better generalization performance, with a higher test f1 score (0.847) compared to Logistic Regression, indicating that nonlinear relationship does provide additional predictive power. Parameter tuning has a noticeable impact on Random Forest performance. The tuned random forest model has the highest test f1 score (0.857) among all the models.

XGBoost performs slightly worse than Random Forest. Also, the tuned XGBoost model doesn't have better performance than the default model, suggesting complexity may not contribute to this dataset.

Based on the test performance, the tuned Random Forest model is selected as the final model for ATP match prediction.

	model	cv_accuracy	cv_f1	test_accuracy	test_f1
0	Logistic Regression	0.900	0.890	0.854	0.841
1	Random Forest	0.869	0.850	0.861	0.847
2	XGBoost	0.859	0.840	0.848	0.835
3	tuned Random Forest	NaN	0.864	0.868	0.857
4	tuned XGBoost	NaN	0.861	0.854	0.841

Table1: ATP model performance metrics

2. WTA dataset

The performance of all machine learning models on the WTA dataset is summarized in the Table2. Logistic Regression, Random Forest, and XGBoost all achieve strong results, with relatively small differences across evaluation metrics (Note: corss-validation accuracy for tuned

models is not reported, as f1 score is used as the primary optimization metric during parameter tuning)

As the baseline model here, Logistic Regression has a solid performance, with a cross-validation f1 score of 0.887 and a test f1-score of 0.909, indicating that linear relationships between features and match outcome are strong enough for the match prediction.

Random Forest has a slightly worse performance than Logistic Regression on both cross-validation (f1 score = 0.884) and test dataset (0.887), indicating that adding nonlinear relationships cannot improve the model performance for this dataset. Tuned Random Forest has the same cross validation f1 score (0.887) as the Logistic Regression, yet the test f1 score (0.881) is lowest of all the models.

XGBoost performs slightly better with test f1 score of 0.899. Yet the tuned XGBoost model has the highest cross validation f1 score of 0.889 and the highest test f1 score of 0.917 of all the models.

Based on the test performance, the tuned XGBoost model is selected as the final model for ATP match prediction.

	model	cv_accuracy	cv_f1	test_accuracy	test_f1
0	Logistic Regression	0.888	0.887	0.913	0.909
1	Random Forest	0.884	0.884	0.893	0.887
2	XGBoost	0.883	0.883	0.900	0.899
3	tuned Random Forest	NaN	0.887	0.887	0.881
4	tuned XGBoost	NaN	0.889	0.920	0.917

Table2: WTA model performance metrics

2.2 Feature Importance Analysis

1. Logistic Regression

The following coefficient plots provide insight into the relative importance and direction of each feature for ATP and WTA separately.

Some common critical findings: winner rate difference is the most significant predictor in both ATP and WTA models. A higher winner rate relative to the opponent will strongly increase the winning probability. Unforced error rate difference is the largest negative feature. Players who commit more unforced errors are significantly less likely to win. This finding highlights the importance of consistency for winning a match. Break point win percentage difference is the second strong feature, which demonstrates the importance of performance under pressure.

The differences between ATP and WTA matches: Net win percentage difference is another important feature particularly in ATP matches, indicating the net play skill is more impactful in the men's game. Return depth (short return) has more negative impact in WTA matches, indicating that return skill is more impactful in the women's game.

Overall speaking, both models show that winners, unforced errors, and break points performance are the dominant factors in determining match outcomes. The ATP model shows stronger contribution from net play, indicating a different game style/strategy preference from WTA. The WTA model shows stronger contribution from break point and return level, indicating that key point execution is more critical for women.

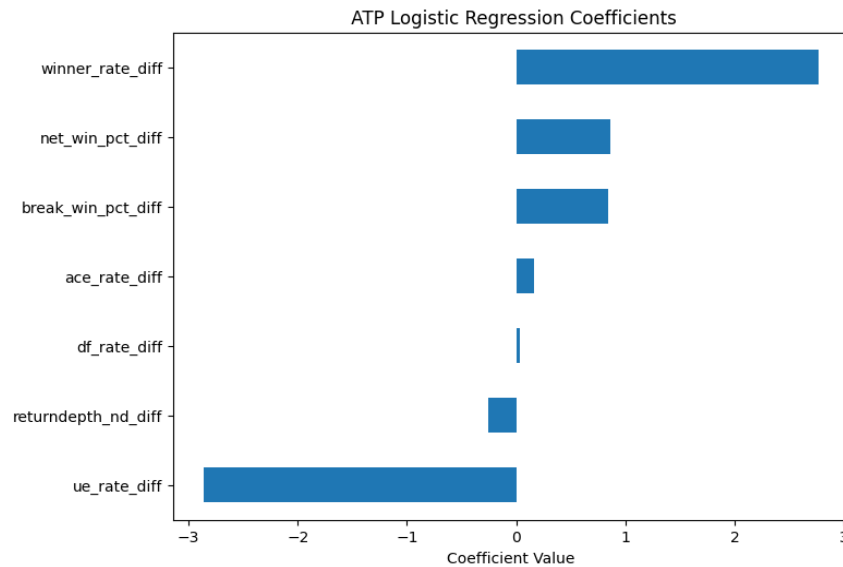


Figure3: ATP coefficients plot

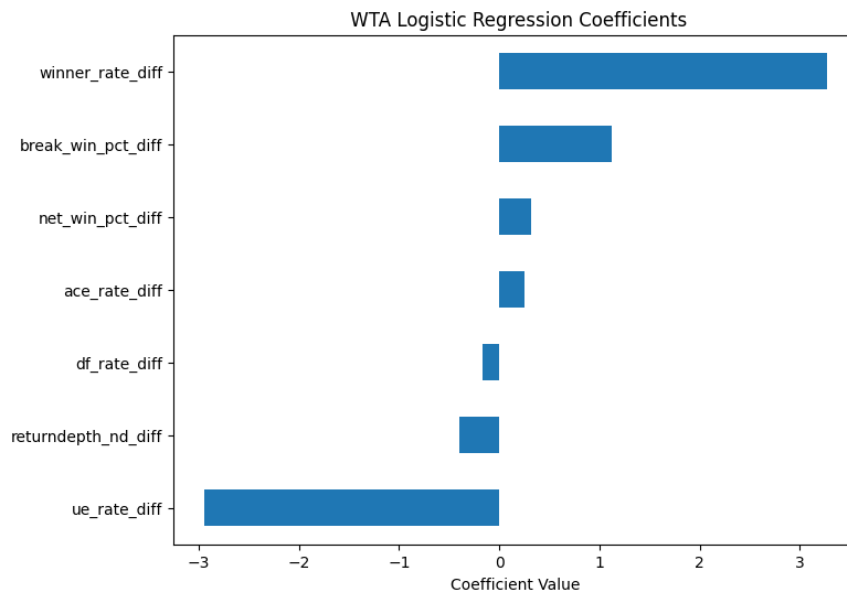


Figure4: WTA coefficients plot

2. Tree-based models (RF/XGB)

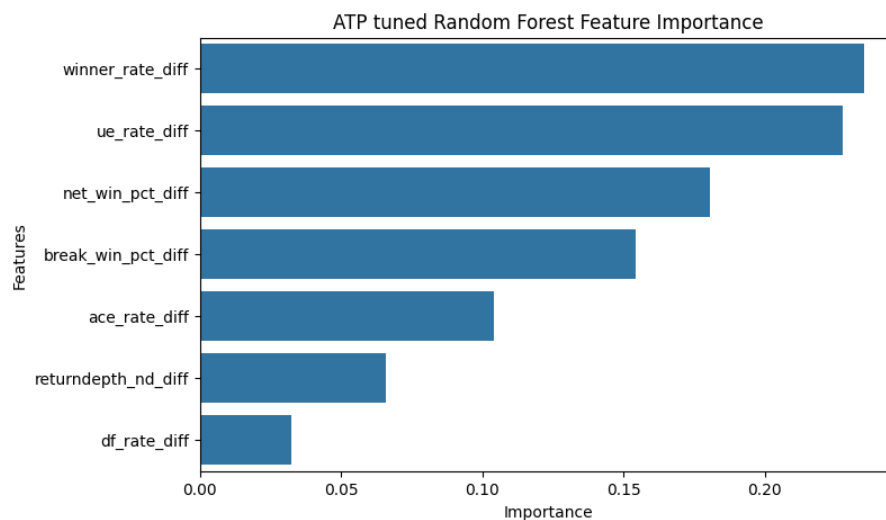


Figure5: ATP feature importance

For the ATP dataset, we use the feature importance of the tuned Radom Forest (the final predictive model).

Some critical results here: winner rate difference is the most important feature; unforced error rate difference is the second most important feature; net win percentage difference and break point win percentage difference also contribute significantly; ace rate difference has moderate importance; return depth has a smaller contribution; double fault rate has the lowest importance.

These findings are consistent with the previously correlation analysis and logistic regression coefficients.

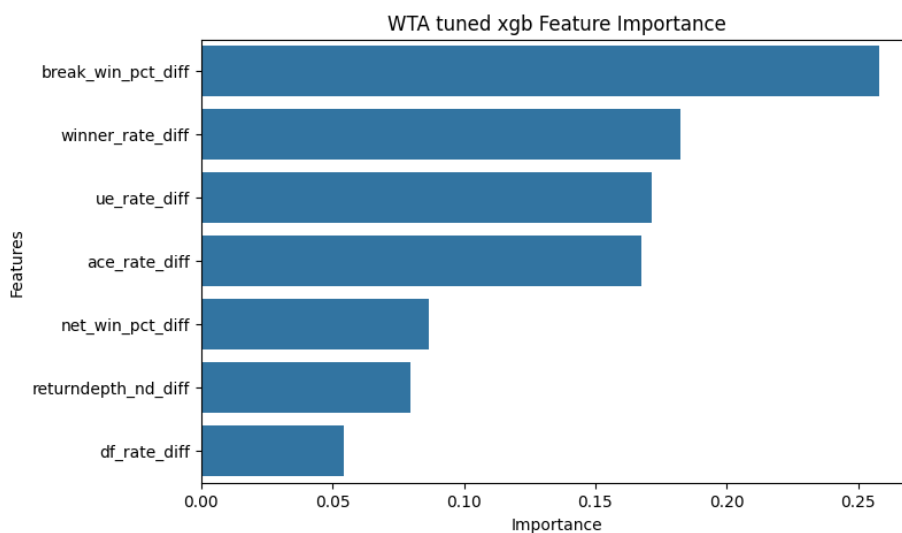


Figure6: WTA feature importance

For the WTA dataset, we use the feature importance of the tuned XGBoost (the final predictive model).

This result shows that break point win percentage difference is the most importance feature; winner rate difference and unforced error rate difference remain highly impactful; ace rate difference shows relatively high importance; net win percentage difference and return depth (return short) have moderate importance; double fault rate difference remains the least important feature.

These findings are mostly consistent with the previously correlation analysis and logistic regression coefficients. The biggest difference is that the break point win percentage difference demonstrates the most significant impact on the match outcome.

Conclusions

1. Insights&recommendations

1. Consistency matters more than power

Across both ATP and WTA analysis, unforced error rate is one of the strongest predictors of match outcome. Players who commit fewer unforced errors relative to their opponents are more likely to win.

For amateur players, reducing errors is more impactful than trying to hit more winners. When in a singles match, focus on keeping the ball in play; avoid unnecessary risk, especially early in rallies; prioritize consistency over hitting aggressive winners.

2. Controlled aggression wins matches

Winner rate is also a top predictor of success. The winning players are not just consistent, they are also able to finish points effectively. The key is to balance avoid attacking anytime, focus on structuring a point to get the final attack move.

For amateur players, we should be patient in rallies, but also recognize opportunities to attack decisively; develop a reliable finishing shot (like a forehand or approach shot); force the opponents to play our rhythm and structure a winner on top of it.

3. Perform better under pressure

The analysis shows that break point performance is the most importance factor in WTA matches and important in ATP matches as well, which demonstrates that tennis is a mental game.

For amateur players, we should play matches/sets more often instead of just rallies to get used to the pressure scenarios; develop a go-to strategy under pressure; stay mentally calm during critical points and adverse scenarios.

4. Net play and tactics are effective tools for winning

Net win percentage contributes meaningfully to match outcome, particularly in ATP matches.

For amateur players, the takeaway here is that tactical variety can disrupt opponents and create point opportunities. We can incorporate net approaches when appropriate, work on volleys and positioning when in practice. It is also a noticeable different pattern in women's and men's tennis game. For women players, be prepared for more net play games if play matches with men's players.

Appendix-A: Data pre-processing engineering

1.1 Point by Point Data Engineering

The definition of the variable 'History' is unknown and cannot be speculated. Speed_KMH is basically the same as Speed_MPH. I removed these two variables from the dataset. I found PointNumber has some string values (eg, 0X, 0Y, 20Y..), which is abnormal. After checking the data distribution, I found that those are extra records and removed them, then converted PointNumber to int64 type.

RallyCount has 35826 values equal to 0. During a point play, the smallest RallyCount is 1, when there is a double fault performed by either one of the players. So, we treat those 0 RallyCount as missing values. First, we replace 0 with NaN, create an ace flag. if there is an ace in a point, the RallyCount can be filled in with 1 confidently. Then we use grouping logic to fill in the remaining non-ace NaNs. We fill in remaining non-ace NaNs with the mean values of RallyCount grouped by SetNo(in which set in point is played) and ServeNumber(0 means double fault, 1 means the first serve, 2 means the second serve). After this, there're still 7709 rows that have NaNs RallyCount. For these rows, we fill in remaining NaNs with the mean values of RallyCount grouped by SetNo.

ServeWidth(serve to the body, wide, or bodywide, center, body center area), ServeDepth(serve deep or not), and ReturnDepth(return deep or not) have 44749, 44749, 77483 missing values separately. These are categorical variables and don't need to be imputed on the point level. We'll get the sum of each different categories of these variables for each of the player on the match level.

1.2 Generating Match Level Data

After the processing of the raw point by point data, we aggregate some of the core columns to get match-level data. The match-level data is defined as 'match_df' and has 49 variables: match_id, points_played, rallies_count, sets_played, p1_points_won, p2_points_won, p1_aces, p2_aces, p1_winners, p2_winners, p1_double_faults, p2_double_faults, p1_unforced_errors, p2_unforced_errors, p1_net_points, p2_net_points, p1_net_points_won, p2_net_points_won, p1_breakpoints, p2_breakpoints, p1_breakpoints_won, p2_breakpoint_won, p1_distance_run, p2_distance_run, max_serve_speed_mph, p1_ServeWidth_B, p1_ServeWidth_BC, p1_ServeWidth_BW, p1_ServeWidth_C, p1_ServeWidth_W, p2_ServeWidth_B, p2_ServeWidth_BC, p2_ServeWidth_BW, p2_ServeWidth_C, p2_ServeWidth_W, p1_ServeDepth_CTL, p1_ServeDepth_NCTL, p2_ServeDepth_CTL, p2_ServeDepth_NCTL, p1_ReturnDepth_D, p1_ReturnDepth_ND, p2_ReturnDepth_D, p2_ReturnDepth_ND, p1_serve_count, p2_serve_count, p1_sets_won, p2_sets_won, match_winner(who wins the match), points_winner(who wins most points in the match; it is highly correlated with match_winner).

1.3. Match Level Data Engineering

Remove rows/matches that has sets_played ==1, since for a match the least number of sets played is 2. Remove matches that has < 48 points_played. Since for a match the least number of points played is 48. After checking the data distribution, p1_distance_run, p2_distance_run and max_serve_speed_mph has some values equal to 0. This is normal because at point level, but

abnormal at match level. I replace 0 value with NaN and then impute them with the mean of each variable.

Then I compute the percentages stats, and difference stats. Since tennis is a relative performance sport. We can only make predictions based on the difference stats, not using the original absolute stats. Next, I combine the percentage stats, difference stats, core stats of a match, serve stats and absolute stats we already have, generate an ultimate dataset for analysis, which is called 'analysis_df'

Since for this project, match_winner is the target of this project (match_winner = 1 means the player1 won, match_winner= 2 means player 2 won). We map winner labels from (1,2) to (1, 0) for the later binary classification usage.

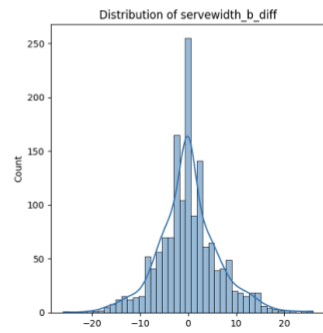
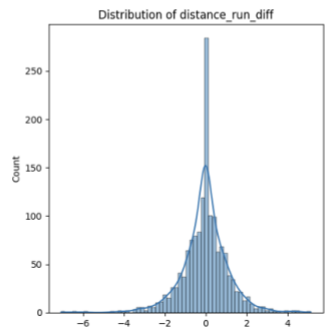
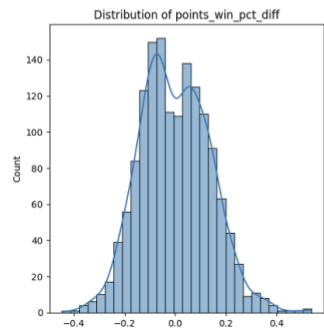
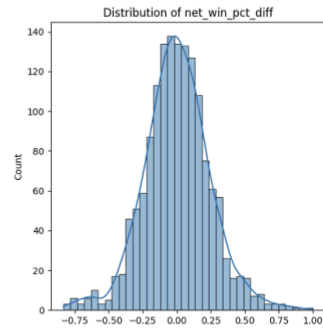
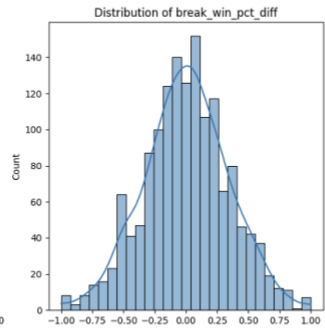
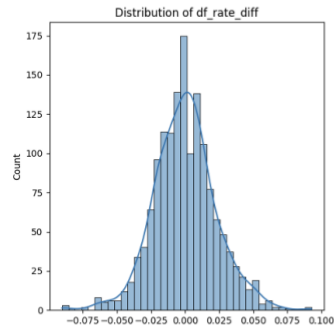
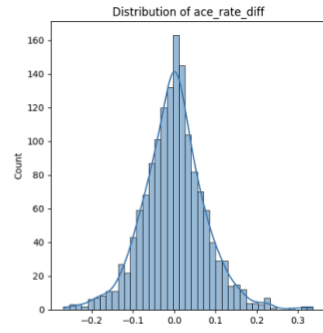
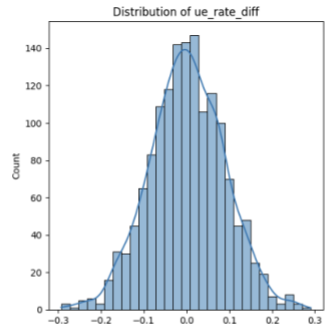
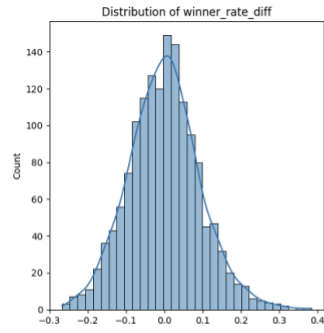
1.4. Final Analysis Dataset Description

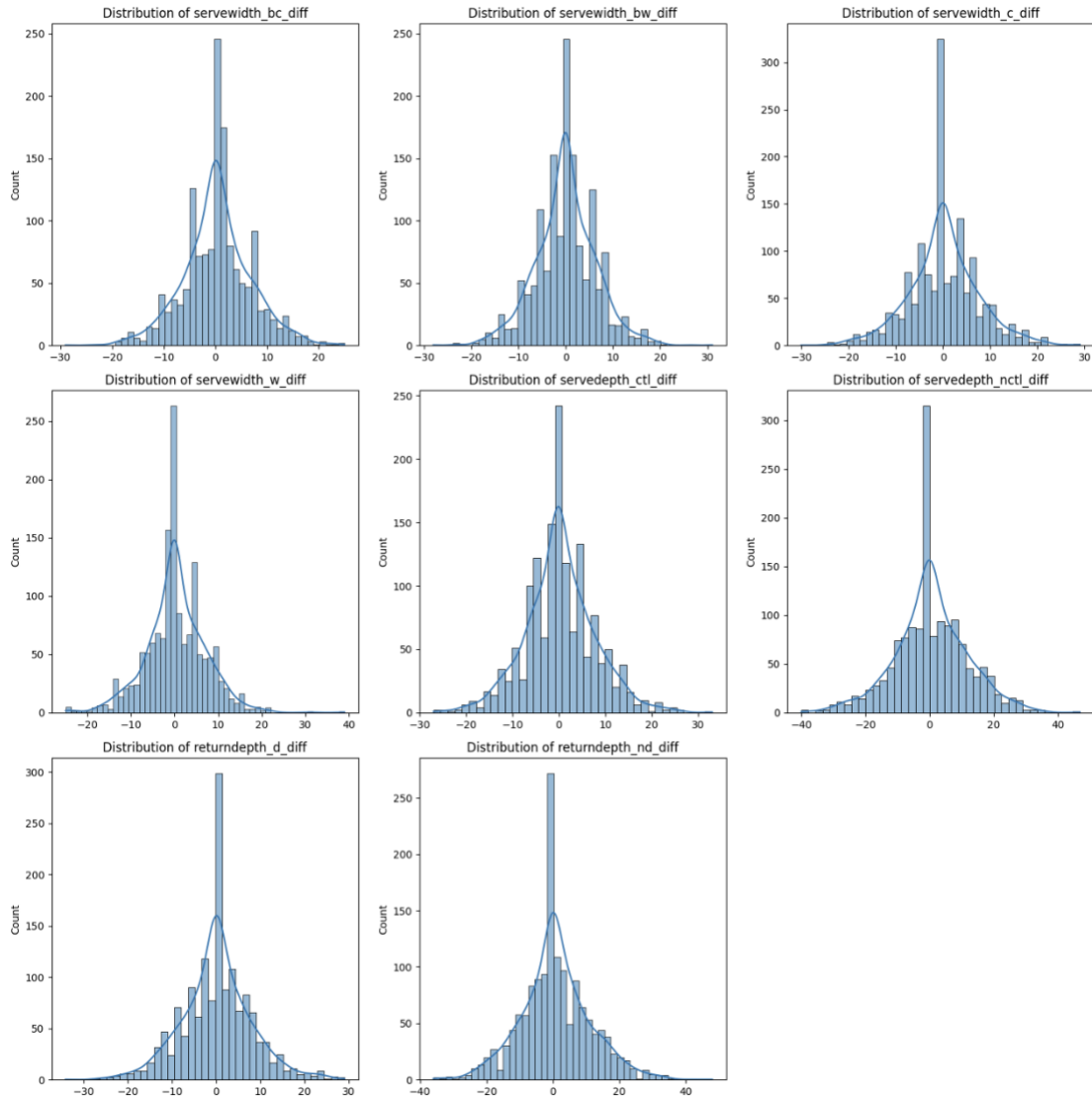
For demonstration and convenient for later analysis, I categorized these data into four types.

1. Absolute variables: 'p1_avg_run_distance', 'p2_avg_run_distance', 'p1_net_win_pct', 'p2_net_win_pct', 'p1_break_win_pct', 'p2_break_win_pct', 'p1_ace_rate', 'p2_ace_rate', 'p1_df_rate', 'p2_df_rate', 'p1_winner_rate', 'p2_winner_rate', 'p1_ue_rate', 'p2_ue_rate', 'p1_points_win_pct', 'p2_points_win_pct'
2. Serve related absolute variables: 'p1_ServeWidth_B', 'p2_ServeWidth_B', 'p1_ServeWidth_BC', 'p2_ServeWidth_BC', 'p1_ServeWidth_BW', 'p2_ServeWidth_BW', 'p1_ServeWidth_C', 'p2_ServeWidth_C', 'p1_ServeWidth_W', 'p2_ServeWidth_W', 'p1_ServeDepth_CTL', 'p2_ServeDepth_CTL', 'p1_ServeDepth_NCTL', 'p2_ServeDepth_NCTL', 'p1_ReturnDepth_D', 'p2_ReturnDepth_D', 'p1_ReturnDepth_ND', 'p2_ReturnDepth_ND'
3. Match info variables: 'match_id', 'points_played', 'sets_played', 'rallies_count', 'max_serve_speed_mph', 'avg_rally_count', 'match_winner', 'points_winner'
4. Performance difference variables: 'winner_rate_diff', 'ue_rate_diff', 'ace_rate_diff', 'df_rate_diff', 'break_win_pct_diff', 'net_win_pct_diff', 'points_win_pct_diff', 'distance_run_diff', 'servewidth_b_diff', 'servewidth_bc_diff', 'servewidth_bw_diff', 'servewidth_c_diff', 'servewidth_w_diff', 'servedepth_ctl_diff', 'servedepth_nctl_diff', 'returndepth_d_diff', 'returndepth_nd_diff'

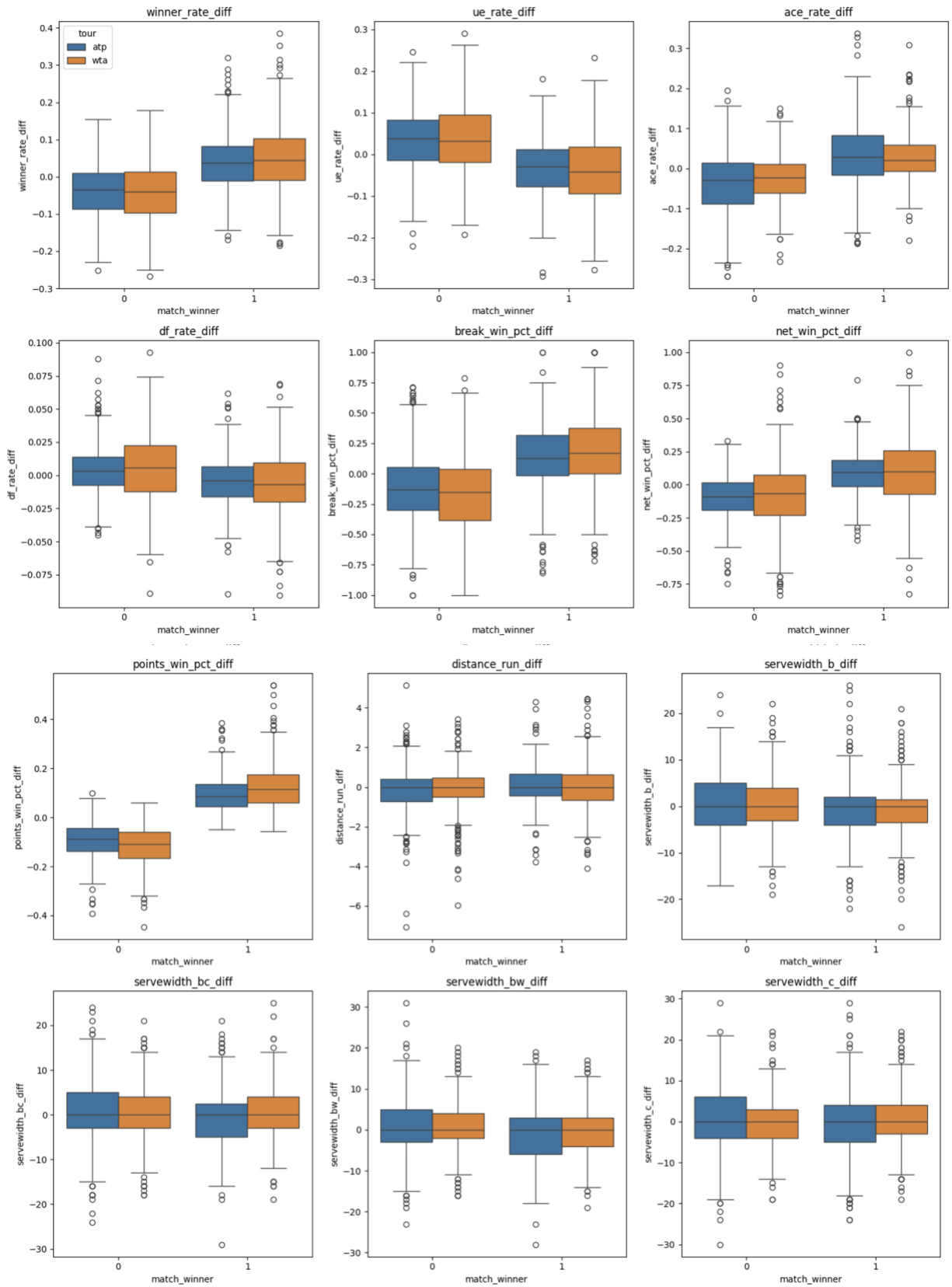
Appendix-B: Supplement plots and tables

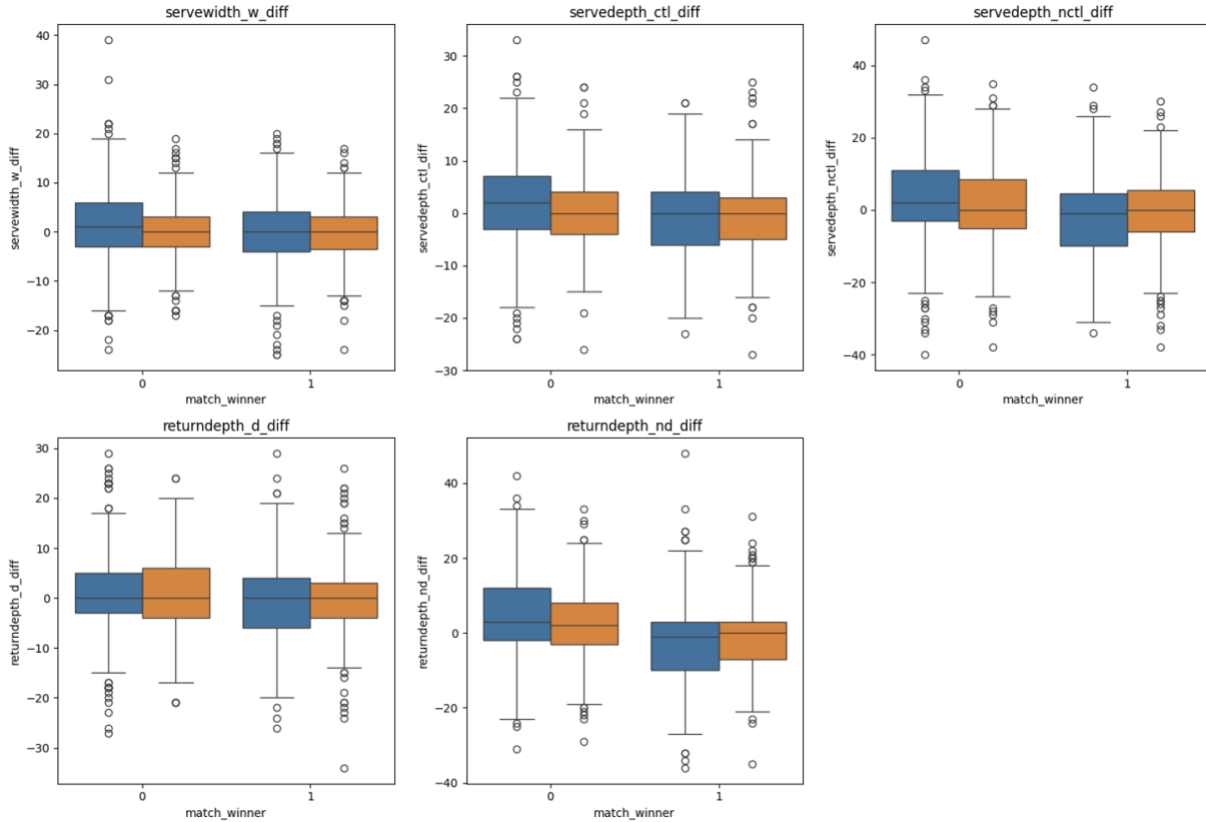
- 1.The distributions of difference-based performance





2. box plot of difference-based performance for both ATP and WTA





3. ATP Features ranked by correlation with match_winner

index	feature	correlation	abs_correlation
0	points_win_pct_diff	0.8006666165777319	0.8006666165777319
1	net_win_pct_diff	0.48439239226548597	0.48439239226548597
2	winner_rate_diff	0.46559334523816776	0.46559334523816776
3	ue_rate_diff	-0.43134427086846305	0.43134427086846305
4	break_win_pct_diff	0.4200726145197232	0.4200726145197232
5	ace_rate_diff	0.406541946604558	0.406541946604558
6	returndepth_nd_diff	-0.2808350644968772	0.2808350644968772
7	servedepth_nctl_diff	-0.22655095371582173	0.22655095371582173
8	df_rate_diff	-0.19822013944596967	0.19822013944596967
9	servedepth_ctl_diff	-0.15039056277640694	0.15039056277640694
10	servewidth_bw_diff	-0.1430306976371941	0.1430306976371941
11	distance_run_diff	0.13705841146772454	0.13705841146772454
12	servewidth_w_diff	-0.11569097751861666	0.11569097751861666

index	feature	correlation	abs_correlation
13	servewidth_bc_diff	-0.1062102017481571	0.1062102017481571
14	returndepth_d_diff	-0.09659409959608212	0.09659409959608212
15	servewidth_c_diff	-0.09557589427996091	0.09557589427996091
16	servewidth_b_diff	-0.08434794801585162	0.08434794801585162
17	max_serve_speed_mph	-0.07873081394050808	0.07873081394050808
18	avg_rally_count	-0.07557223922878789	0.07557223922878789
19	sets_played	0.04277686481879476	0.04277686481879476
20	points_played	0.03769703286210631	0.03769703286210631
21	rallies_count	-0.030141617905789862	0.030141617905789862

4. Features ranked by correlation with match_winner

index	feature	correlation	abs_correlation
0	points_win_pct_diff	0.8115421912448822	0.8115421912448822
1	break_win_pct_diff	0.4796224504419589	0.4796224504419589
2	winner_rate_diff	0.4611704436947489	0.4611704436947489
3	ace_rate_diff	0.4236798823707106	0.4236798823707106
4	ue_rate_diff	-0.4143250848129913	0.4143250848129913
5	net_win_pct_diff	0.3046592445459361	0.3046592445459361
6	df_rate_diff	-0.21846509727583707	0.21846509727583707
7	returndepth_nd_diff	-0.20177253073985546	0.20177253073985546
8	servewidth_b_diff	-0.12973646179123002	0.12973646179123002
9	servedepth_nctl_diff	-0.11189880163399424	0.11189880163399424
10	servewidth_bw_diff	-0.10556418160527867	0.10556418160527867
11	returndepth_d_diff	-0.0775447174369645	0.0775447174369645
12	servewidth_w_diff	-0.055079680150056806	0.055079680150056806
13	servedepth_ctl_diff	-0.0547487728397633	0.0547487728397633
14	servewidth_c_diff	0.04523786347424711	0.04523786347424711
15	distance_run_diff	0.029573396547526224	0.029573396547526224
16	max_serve_speed_mph	0.027408984635014524	0.027408984635014524
17	sets_played	0.026802282801198323	0.026802282801198323
18	servewidth_bc_diff	-0.02445917171607464	0.02445917171607464

index	feature	correlation	abs_correlation
19	points_played	0.01782706933418851	0.01782706933418851
20	rallies_count	0.011963465929020005	0.011963465929020005
21	avg_rally_count	-0.00029938870178999514	0.00029938870178999514